

$$\begin{array}{ccccc}
F([X, Y]c) & \xrightarrow{F\Gamma_{X,Y}^Z} & F[[Z, X]c, [Z, Y]c]c & \xrightarrow{\varphi_{[Z,X],[Z,Y]}} & [F[Z, X]c, F[Z, Y]c]_{\mathcal{D}} \\
\varphi_{X,Y} \downarrow & & & & \downarrow [\text{id}, \varphi_{Z,Y}] \\
[FX, FY]_{\mathcal{D}} & \xrightarrow{\Gamma_{FX,FY}^{FZ}} & [[FZ, FX]_{\mathcal{D}}, [FZ, FY]_{\mathcal{D}}]_{\mathcal{D}} & \xrightarrow{[\varphi_{Z,X}, \text{id}]} & [F[Z, X]c, [FZ, FY]_{\mathcal{D}}]_{\mathcal{D}}
\end{array}$$